

A counterexample to convexity of Lorentz-entanglement annihilating maps

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Source and question

The source paper is Francesca La Piana and Alexander Mueller-Hermes, *Annihilating and breaking Lorentz cone entanglement*, arXiv:2506.14480 [1]. On page 3, after defining the Lorentzian tensor product and the class $\text{LorEA}_2(\mathcal{C}_A, \mathcal{C}_B)$, the authors write:

“The set of these maps will be denoted by $\text{LorEA}_2(\mathcal{C}_A, \mathcal{C}_B)$, and we do not know whether it is convex in general.”

They then prove convexity in the special case where $\mathcal{C}_B = \mathcal{L}_n$ is a Lorentz cone.

Definition 1.1 (Lorentzian tensor product). *For proper cones $\mathcal{C}_A$ and $\mathcal{C}_B$ we define the Lorentzian tensor product as*

$$\mathcal{C}_A \otimes_L \mathcal{C}_B := \text{conv}\{(A \otimes B) \left(\hat{I}_k \right) : A \in \text{Pos}(\mathcal{L}_k, \mathcal{C}_A), B \in \text{Pos}(\mathcal{L}_k, \mathcal{C}_B), k \in \mathbb{N}\},$$

where $\hat{I}_k = \sum_{i=0}^k e_i \otimes e_i$ denotes the identity tensor in $\mathcal{L}_k \otimes_{\max} \mathcal{L}_k$.

The Lorentzian tensor product defines a tensor product in the category of ordered vector spaces since

$$\mathcal{C}_A \otimes_{\min} \mathcal{C}_B \subseteq \mathcal{C}_A \otimes_L \mathcal{C}_B \subseteq \mathcal{C}_A \otimes_{\max} \mathcal{C}_B.$$

It associates to any pair of proper cones $\mathcal{C}_A \subseteq \mathcal{V}_A$ and $\mathcal{C}_B \subseteq \mathcal{V}_B$ a reasonable cross cone (see [dB20]) and it is an example of a tensor cone (see [dB20, Section 8]), i.e., the ordered vector space analogue of a tensor norm in the category of Banach spaces. Like similar constructions for Banach spaces, the Lorentzian tensor product is not associative (see Appendix C).

Similar to the 2-max-entanglement annihilating maps introduced above, we can consider linear maps annihilating the entanglement described by the Lorentzian tensor product. A linear map $P \in \text{Pos}(\mathcal{C}_A, \mathcal{C}_B)$ is called *Lorentz-entanglement annihilating* if

$$(P \otimes P)(\mathcal{C}_A \otimes_L \mathcal{C}_A) \subseteq \mathcal{C}_B \otimes_{\min} \mathcal{C}_B.$$

The set of these maps will be denoted by $\text{LorEA}_2(\mathcal{C}_A, \mathcal{C}_B)$, and we do not know whether it is convex in general. However, in the special case where $\mathcal{C}_B = \mathcal{L}_n$ is a Lorentz cone we show that the set $\text{LorEA}_2(\mathcal{C}_A, \mathcal{L}_n)$ is a convex cone.

Definitions

All cones are finite-dimensional proper cones. For proper cones $\mathcal{C}, \mathcal{D}$, write

$$\text{maxEA}_2(\mathcal{C}, \mathcal{D}) = \{T \in \text{Pos}(\mathcal{C}, \mathcal{D}) : (T \otimes T)(\mathcal{C} \otimes_{\max} \mathcal{C}) \subseteq \mathcal{D} \otimes_{\min} \mathcal{D}\}.$$

The Lorentzian tensor product is

$$\mathcal{C} \otimes_L \mathcal{D} = \text{conv}\{(A \otimes B)(\widehat{I}_k) : A \in \text{Pos}(\mathcal{L}_k, \mathcal{C}), B \in \text{Pos}(\mathcal{L}_k, \mathcal{D}), k \in \mathbb{N}\},$$

and

$$\text{LorEA}_2(\mathcal{C}, \mathcal{D}) = \{T \in \text{Pos}(\mathcal{C}, \mathcal{D}) : (T \otimes T)(\mathcal{C} \otimes_L \mathcal{C}) \subseteq \mathcal{D} \otimes_{\min} \mathcal{D}\}.$$

For a finite-dimensional normed space X , let $\mathcal{C}_X = \{(t, x) \in \mathbb{R} \times X : t \geq \|x\|\}$. We use the standard self-duality of $\mathcal{L}_n$ and the identification $(\mathcal{C}_X)^* = \mathcal{C}_{X^*}$.

Proof intuition

The obstruction is already present in the source paper's appendix, but in the wrong orientation for the question as stated there. The appendix gives two maps $P_1, P_2 : \mathcal{C}_{\ell_\infty^2} \rightarrow \mathcal{L}_2$ that are individually 2-max-entanglement annihilating, while $P_1 + P_2$ is not. Taking trace adjoints reverses the cones and preserves the max-annihilation property, so the adjoints $P_1^*, P_2^* : \mathcal{L}_2 \rightarrow \mathcal{C}_{\ell_1^2}$ are again individually max-annihilating and have a non-annihilating sum. Since the input cone is Lorentz, the Lorentzian and maximal tensor products agree on the input side. Thus the same adjoints are a counterexample to convexity of $\text{LorEA}_2(\mathcal{L}_2, \mathcal{C}_{\ell_1^2})$.

Lemma 1 (Lorentz input reduces LorEA to maxEA). *For every proper cone $\mathcal{D}$ and every n ,*

$$\mathcal{L}_n \otimes_L \mathcal{D} = \mathcal{L}_n \otimes_{\max} \mathcal{D}.$$

Consequently

$$\text{LorEA}_2(\mathcal{L}_n, \mathcal{D}) = \text{maxEA}_2(\mathcal{L}_n, \mathcal{D}).$$

Proof. The inclusion $\otimes_L \subseteq \otimes_{\max}$ is part of the general construction. Conversely, identify $\mathcal{L}_n$ with its dual. A tensor $z \in \mathcal{L}_n \otimes_{\max} \mathcal{D}$ corresponds, by the Choi-Jamiołkowski identification for cones, to a positive map $B : \mathcal{L}_n \rightarrow \mathcal{D}$, and

$$z = (\text{id}_{\mathcal{L}_n} \otimes B)(\widehat{I}_n).$$

This tensor is one of the generators in the definition of $\mathcal{L}_n \otimes_L \mathcal{D}$, with $A = \text{id}_{\mathcal{L}_n}$. Hence the two tensor products coincide. Applying this equality to the input tensor product $\mathcal{L}_n \otimes_L \mathcal{L}_n$ gives the stated equality of map classes. $\square$

Lemma 2 (Adjoint duality for max-annihilation). *Let $T : \text{span } \mathcal{C} \rightarrow \text{span } \mathcal{D}$ be linear. Then*

$$T \in \text{maxEA}_2(\mathcal{C}, \mathcal{D}) \iff T^* \in \text{maxEA}_2(\mathcal{D}^*, \mathcal{C}^*).$$

Proof. The equivalence of positivity of T and T^* is standard. The remaining condition is the dual of a cone inclusion. Indeed,

$$(T \otimes T)(\mathcal{C} \otimes_{\max} \mathcal{C}) \subseteq \mathcal{D} \otimes_{\min} \mathcal{D}$$

holds if and only if

$$(T^* \otimes T^*)((\mathcal{D} \otimes_{\min} \mathcal{D})^*) \subseteq (\mathcal{C} \otimes_{\max} \mathcal{C})^*.$$

Using

$$(\mathcal{D} \otimes_{\min} \mathcal{D})^* = \mathcal{D}^* \otimes_{\max} \mathcal{D}^*, \quad (\mathcal{C} \otimes_{\max} \mathcal{C})^* = \mathcal{C}^* \otimes_{\min} \mathcal{C}^*$$

gives precisely $T^* \in \text{maxEA}_2(\mathcal{D}^*, \mathcal{C}^*)$. $\square$

Theorem 1. *The set $\text{LorEA}_2(\mathcal{L}_2, \mathcal{C}_{\ell_1^2})$ is not convex.*

Proof. Let $X = \ell_\infty^2$, so $X^* = \ell_1^2$. Consider the two maps $\phi^{(1)}, \phi^{(2)} : X \rightarrow \ell_2^2$ from Appendix B of [1]:

$$\phi^{(1)} = \begin{pmatrix} 1 & 0 \\ 0 & 1/5 \end{pmatrix}, \quad \phi^{(2)} = \begin{pmatrix} 1 & 3/10 \\ 0 & 1/2 \end{pmatrix}.$$

For such ϕ , the source paper defines

$$\alpha(\phi) = \max \left\{ \|\phi_1 + \phi_2\|_2, \|\phi_1 - \phi_2\|_2, \sqrt{\|\phi H \phi^T\|_1} : H \in \mathcal{H} \right\},$$

where

$$\mathcal{H} = \left\{ \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right\}.$$

Their Appendix B criterion says that

$$\lambda \oplus \phi \in \max\text{EA}_2(\mathcal{C}_{\ell_\infty^2}, \mathcal{L}_2) \iff \alpha(\phi) \leq \lambda.$$

The displayed matrices satisfy

$$\alpha(\phi^{(1)}) + \alpha(\phi^{(2)}) < \alpha(\phi^{(1)} + \phi^{(2)}).$$

For example, exact arithmetic gives

$$\alpha(\phi^{(1)}) = \frac{2^{3/4} 97^{1/4}}{5} \approx 1.055590386, \quad \alpha(\phi^{(2)}) = \frac{\sqrt{194}}{10} \approx 1.392838828,$$

while

$$\alpha(\phi^{(1)} + \phi^{(2)}) = \frac{\sqrt{14} 1889^{1/4}}{10} \approx 2.466732225.$$

Hence, with $P_i = \alpha(\phi^{(i)}) \oplus \phi^{(i)}$, the two maps P_1, P_2 belong to $\max\text{EA}_2(\mathcal{C}_{\ell_\infty^2}, \mathcal{L}_2)$, but

$$P_1 + P_2 = (\alpha(\phi^{(1)}) + \alpha(\phi^{(2)})) \oplus (\phi^{(1)} + \phi^{(2)}) \notin \max\text{EA}_2(\mathcal{C}_{\ell_\infty^2}, \mathcal{L}_2).$$

Set $Q_i = P_i^* : \mathcal{L}_2 \rightarrow (\mathcal{C}_{\ell_\infty^2})^* = \mathcal{C}_{\ell_1^2}$. By Lemma 2,

$$Q_i \in \max\text{EA}_2(\mathcal{L}_2, \mathcal{C}_{\ell_1^2}) \quad (i = 1, 2),$$

and by Lemma 1 this is the same as

$$Q_i \in \text{LorEA}_2(\mathcal{L}_2, \mathcal{C}_{\ell_1^2}).$$

However,

$$Q_1 + Q_2 = (P_1 + P_2)^* \notin \max\text{EA}_2(\mathcal{L}_2, \mathcal{C}_{\ell_1^2}) = \text{LorEA}_2(\mathcal{L}_2, \mathcal{C}_{\ell_1^2}),$$

again by Lemmas 2 and 1. Since this class is closed under positive scalar multiplication, the midpoint $(Q_1 + Q_2)/2$ also fails to belong to it. Therefore $\text{LorEA}_2(\mathcal{L}_2, \mathcal{C}_{\ell_1^2})$ is not convex. $\square$

Verification notes

The accompanying script `code/verify_constants.py` computes the six terms in α for $\phi^{(1)}$, $\phi^{(2)}$, and $\phi^{(1)} + \phi^{(2)}$, and verifies the strict inequality. The proof uses the source paper's Appendix B criterion, trace-duality of minimal and maximal tensor products, and the equality $\mathcal{L}_n \otimes_L \mathcal{D} = \mathcal{L}_n \otimes_{\max} \mathcal{D}$ for Lorentz input.

Novelty check and limitations

On 2026-06-20, I searched the web for the exact phrases

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"LorEA_2" convex "Lorentz-entanglement annihilating",  
"Lorentz-entanglement annihilating" "convex",
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and for the source title together with `convex`. I found the source arXiv paper but no later paper or note resolving the general convexity question. The argument is intentionally narrow: it disproves general convexity, but it does not classify which output cones still give convex Lorentz-entanglement annihilating classes.

Human review recommendation

This packet should be reviewed as a likely valid counterexample. The key points to check are Lemma 2, the Lorentz-input equality in Lemma 1, and the use of the appendix criterion from the source paper with output $\mathcal{L}_2$.

References

- [1] Francesca La Piana and Alexander Mueller-Hermes, *Annihilating and breaking Lorentz cone entanglement*, arXiv:2506.14480, 2025.